GEOMETRIC INTELLIGENCE ENGINE (GIE)

Milk Hill 2001 - Spiral-Arm Boundary / Five-Bit Phase Analysis

Working analytical record - 16 August 2026

1. Scope and controlling geometry

This record documents the current geometry-defined decoding work through the six-arm boundary test. It does not alter the accepted GIE reconstruction or claim an English-language message that has not yet been reproduced.

Controlling quantity	Value
Overall formation diameter	787 ft
Largest circle diameter	72 ft
Smallest circle diameter	40 in
Total circles	409
Structure	1 center + 6 arms x 68 circles
Arm separation	60 degrees

2. Binary extraction

Within each arm, the binary rule is: bit = 1 when the current circle diameter is greater than the previous circle diameter; otherwise bit = 0. With 68 circles on each arm, each arm contributes 67 diameter-to-diameter comparisons. Across six arms this yields 6 x 67 = 402 binary comparisons.

The 402-bit stream contains 80 complete five-bit groups (400 bits) plus 2 trailing bits.

3. Direct five-bit output

Direct output:
SFKBOJHWXFQIGLATVV<29>YCRYSLPAIL<27>JKIKNLEST<28><27>JFARWKP<27><28>QKJIDQOEAIDISIJI<28>ZVGNYJ<27>IML<28> E

4. The arm-boundary phase result

The key structural result is arithmetic, not interpretive: 67 is not divisible by 5.

67 = 13 x 5 + 2

Therefore, if five-bit grouping is kept continuous while traversal moves from one 68-circle arm to the next, the five-bit frame advances by 2 bits at every arm boundary. The starting phase evolves:

0 -> 2 -> 4 -> 1 -> 3 -> 0

5. Five-bit groups that actually cross arm boundaries

Boundary	Global bits	Five bits	Value	Interpretation
Arm 0 -> Arm 1	66-70	01100	12	L
Arm 1 -> Arm 2	131-135	00001	1	A
Arm 2 -> Arm 3	201-205	11011	27	CONTROL 27
Arm 3 -> Arm 4	266-270	01001	9	I
Arm 4 -> Arm 5	335/336	Boundary falls exactly between groups	-	No cross-boundary group

This shows that some five-bit symbols are literally composed from bits on two neighboring spiral arms. That makes the arm boundary a mathematically real part of the decoding problem.

6. Per-arm reset check

As a control test, each arm was also decoded independently by restarting five-bit grouping at the beginning of that arm. Each arm produces 13 full groups plus a 2-bit remainder:

Arm	13-group output	2-bit remainder
0	SFKBOJHWXFQIG	01
1	PFRZ<27>WDNKFMR SPACE	00
2	TVMUETUWFBYZN	01
3	VTLCENWAWYBVT	10
4	IDKYHJIBLZJRO	00
5	ZVGNYJ<27>IML<28> SPACE E	10

7. Between-arm / radial-column test

To test the specific idea that information may be defined between neighboring spiral arms, the corresponding radial comparison at each rank was examined across all six arms. Direct six-bit columns, adjacent-arm XOR, and adjacent-arm equality/complement variants were tested. Under the existing alphabet/control mapping, these operations did not produce a reproducible English sentence.

That negative result is useful: it rules out several simple between-arm interpretations without changing the geometry or forcing language into the output.

8. Current result

- The 409-circle, six-arm geometry remains the controlling reconstruction.
- The binary extraction rule is reproducible and yields 402 comparisons.
- The direct five-bit stream is reproducible.
- The 2-bit phase advance at each 67-comparison arm boundary is an exact structural consequence.
- An English-language message has not yet been established by the tested boundary/radial-column rules.
- The next justified decoding work is to determine whether CONTROL values 27-31 have a geometry-defined role in handling arm-boundary phase/carry, rather than treating them as ordinary letters or arbitrarily rearranging output.

9. What Milk Hill shares with the world

What can be shared responsibly today is not a claimed sentence but a remarkable mathematical object: a large sixfold spiral that can be represented as a precise count, scale, radial ordering, binary comparison sequence, and phase structure. Its value is that it invites a reproducible question: can a visual geometric arrangement carry information under fixed rules?

The strongest public lesson is methodological. Geometry, measurement, binary encoding, and language must remain separate stages. A meaningful decode must survive a fixed rule without changing the rule after seeing the output. Whether or not the final stage becomes English, the formation already provides a concrete example of symmetry, sequence, information representation, and the difference between discovering a pattern and proving an interpretation.

10. Next test

Continue from the exact 2-bit arm-boundary phase shift. Test control values 27-31 specifically as possible boundary/carry instructions using only rules determined from the geometry and binary stream before reading resulting language. Preserve the existing direct stream and record every tested rule.

11. Control-code phase test

A constrained follow-up test treated five-bit values 27-31 as possible phase instructions because there are exactly five non-letter control values and five possible five-bit rotations. The natural mappings 27-31 -> phase positions 0-4 and 1-5 were tested as set, cumulative-add, and cumulative-subtract rules, with both left and right five-bit rotation. A parallel test treated the same values as small Caesar-style shifts on subsequent A-Z symbols.

None of these simple control-as-phase rules produced a stable English-language output. This means the control values cannot presently be assigned those meanings without adding an unsupported rule.

12. Direct two-arm carry symbols

A more geometry-specific boundary test was then performed. Each arm has 67 comparison bits = 13 complete five-bit groups + a two-bit remainder. For every neighboring arm pair, the two remainder bits from the first arm were joined to the first three comparison bits of the next arm. This creates exactly one five-bit symbol located at each arm-to-arm boundary. The sixth boundary closes Arm 5 back to Arm 0, respecting the sixfold cyclic arrangement.

Boundary	Carry 2	Next 3	Five bits	Value	Symbol
Arm 0 -> Arm 1	01	100	01100	12	L
Arm 1 -> Arm 2	00	101	00101	5	E
Arm 2 -> Arm 3	01	101	01101	13	M
Arm 3 -> Arm 4	10	010	10010	18	R
Arm 4 -> Arm 5	00	110	00110	6	F
Arm 5 -> Arm 0	10	100	10100	20	T

Boundary-symbol sequence: L E M R F T

This sequence is a real result of the specific two-arm carry construction. It is not yet an English word or sentence. No arbitrary rearrangement is accepted as a decode. The value of the result is that it produces one reproducible symbol for each of the six physical interfaces between neighboring arms.

13. Orientation controls

The same boundary construction was checked under the small set of geometry-defined orientation alternatives: clockwise versus counterclockwise arm order, outward versus inward traversal, and carry-bits-first versus next-arm-bits-first. The resulting six-symbol strings were:

- Clockwise / outward / carry first: LEMRFT
- Clockwise / outward / next first: QTUJXR
- Clockwise / inward / carry first: HLJIS<29>
- Clockwise / inward / next first: AQIENW
- Counterclockwise / outward / carry first: NREULD
- Counterclockwise / outward / next first: YJTVQP
- Counterclockwise / inward / carry first: KYRLHM
- Counterclockwise / inward / next first: MGJQAU

None is accepted as a linguistic decode. Several contain ordinary letter-like fragments by chance, but there is no geometry-defined reason to reorder their letters into a desired word.

14. Updated stopping point

The strongest new result is therefore not a sentence but a reproducible boundary alphabet candidate: six interfaces can each generate a five-bit symbol by carrying the two-bit remainder of one arm into the first three bits of its neighbor. The canonical clockwise/outward construction yields LEMRFT. Future work should test whether those six boundary symbols act as ordering keys, delimiters, or controls for the 13 complete five-bit groups inside each arm. Any such rule must be derived from the geometry before judging the resulting language.